

Spotify Top 50 World Music Analytics

Project Documentation
Interactive Power BI Dashboard | DAX | Data Visualization

Project	Spotify Top 50 World Music Analytics
Tool	Microsoft Power BI
Analytics	DAX + Power Query
Dataset	Spotify Top 50 World Music Dataset
Author	Chanchal Khandkure
Role	Aspiring Data Analyst

Purpose: Transform Spotify Top 50 chart data into an interactive dashboard for analyzing song popularity, artist performance, chart rankings, explicit content, duration, albums and release trends.

1. Project Overview

This project analyzes Spotify Top 50 music data using Microsoft Power BI. The dashboard converts raw chart and song attributes into interactive KPIs, rankings, artist analysis, popularity categories and time-based insights.

2. Project Objectives

- Analyze Spotify Top 50 songs and artists.
- Measure chart performance using position, Top 10, Top 20 and #1 appearances.
- Understand the distribution of Spotify popularity scores.
- Compare explicit and non-explicit songs.
- Analyze song duration in minutes.
- Explore album type and track-count information.
- Analyze chart and release dates over time.
- Build an interactive, portfolio-ready Power BI dashboard.
- Demonstrate practical DAX and Power BI skills.

3. Dataset Structure

Column	Description
date	Chart date
position	Song chart position
song	Song name
artist	Artist name
popularity	Spotify popularity score
duration_ms	Song duration in milliseconds
album_type	Album type
total_tracks	Number of tracks in album
release_date	Song/album release date
is_explicit	Explicit-content indicator
album_cover_url	Album cover URL

4. Dashboard Pages

Overview Dashboard

- Total Songs
- Total Artists
- Average Popularity
- Best/Worst Position
- Position 1 Hits
- Explicit Song %
- Average Duration
- Top songs and artists
- Explicit vs non-explicit analysis

Artist Analysis

- Top artists by appearances
- Top 10 performance
- #1 hits
- Average popularity by artist
- Best position by artist
- Songs per artist

Song Analysis

- Top songs by popularity
- Chart ranking
- Top 10/20/50 analysis
- Duration
- Explicit content
- Album information

5. Data Preparation

- Checked dataset structure and data types.
- Prepared date fields for chart and release analysis.
- Used duration_ms and converted milliseconds to minutes in DAX.
- Created calculated columns for Year, Month, Month Number, Release Year and Release Month.
- Created DAX measures for KPIs, rankings, percentages and analytical categories.
- Prepared fields for Power BI visualizations.

6. DAX Calculations

Total Songs

```
Total Songs =  
DISTINCTCOUNT('Top-50-World'[song])
```

Total Artists

```
Total Artists =  
DISTINCTCOUNT('Top-50-World'[artist])
```

Average Popularity

```
Average Popularity =  
AVERAGE('Top-50-World'[popularity])
```

Position 1 Hits

```
Position 1 Hits =  
CALCULATE(  
COUNTROWS('Top-50-World'),  
'Top-50-World'[position] = 1  
)
```

Top 10 Appearances

```
Top 10 Appearances =  
CALCULATE(  
COUNTROWS('Top-50-World'),  
'Top-50-World'[position] <= 10  
)
```

Top 20 Appearances

```
Top 20 Appearances =  
CALCULATE(  
COUNTROWS('Top-50-World'),  
'Top-50-World'[position] <= 20  
)
```

Explicit Song %

```
Explicit Song % =  
DIVIDE(  
[Explicit Songs],  
COUNTROWS('Top-50-World'),  
0  
)
```

Average Duration Minutes

```
Average Duration Minutes =  
DIVIDE(  
AVERAGE('Top-50-World'[duration_ms]),  
60000,  
0  
)
```

Highly Popular Songs

```
Highly Popular Songs =  
CALCULATE(  
DISTINCTCOUNT('Top-50-World'[song]),  
'Top-50-World'[popularity] >= 80  
)
```

Average Songs per Artist

```
Average Songs per Artist =  
DIVIDE(  
[Total Songs],  
[Total Artists],  
0  
)
```

Total Records

```
Total Records =  
COUNTROWS('Top-50-World')
```

Median Popularity

```
Median Popularity =  
MEDIAN('Top-50-World'[popularity])
```

7. Calculated Columns

Chart Year

```
Year = YEAR('Top-50-World'[date])
```

Chart Month

```
Month = FORMAT('Top-50-World'[date], "MMM")
```

Month Number

```
Month Number = MONTH('Top-50-World'[date])
```

Release Year

```
Release Year = YEAR('Top-50-World'[release_date])
```

Release Month

```
Release Month = FORMAT('Top-50-World'[release_date], "MMM")
```

Release Month Number

```
Release Month Number = MONTH('Top-50-World'[release_date])
```

8. Popularity Classification

Popularity Score	Category
80+	Highly Popular
60–79	Popular
40–59	Moderate
Below 40	Low

9. Dashboard Design

The dashboard uses a Spotify-inspired dark visual theme with green as the primary accent color.

Element	Hex Code
Background	#121212
Secondary Background	#181818
Card Background	#242424
Spotify Green	#1DB954
Bright Green	#1ED760
Primary Text	#FFFFFF
Secondary Text	#B3B3B3
Border	#303030

10. Visualization Strategy

- KPI Cards – high-level performance metrics.
- Bar Charts – top artists, songs and categories.
- Line Charts – chart and release trends.
- Donut Charts – explicit content, album type and popularity categories.
- Scatter Plots – relationships such as popularity vs duration and popularity vs position.

11. Analytics Workflow

Raw Spotify Dataset → Data Cleaning → Power Query Transformation → Data Modeling → DAX Measures & Calculated Columns → Visualization → Interactive Power BI Dashboard → Music & Artist Insights

12. Key Analytical Findings

- The dashboard provides a structured view of song and artist performance.
- Top 10, Top 20 and #1 metrics provide ranking-based performance analysis.
- Popularity can be analyzed separately from chart position.
- Explicit and non-explicit songs can be compared using count, percentage and popularity measures.
- Duration analysis provides average, minimum, maximum and median song length.
- Album and release-date fields provide additional context for music trends.

13. Skills Demonstrated

Microsoft Power BI • DAX • Power Query • Data Cleaning • Data Transformation • Data Visualization • KPI Development • Ranking Analysis • Artist Analysis • Song Analysis • Date Analysis • Dashboard Design • Data Storytelling

14. Future Improvements

- Spotify API integration
- Dynamic time-intelligence analysis

- Artist profile drill-through pages
- Song-level drill-through
- Year-over-year popularity analysis
- Popularity trend forecasting
- Genre-level analysis
- Interactive Spotify song links
- Machine-learning-based popularity prediction

15. Project Structure

```
Spotify-Dashboard/  
■■■ README.md  
■■■ Spotify_Analysis.pbix  
■■■ spotify-top-50-world.csv  
■■■ Screenshots/  
■■■ Artists.png  
■■■ Index.png  
■■■ Overview.png  
■■■ Songs.png
```

16. Conclusion

This project demonstrates how raw Spotify Top 50 data can be transformed into an interactive Power BI dashboard using data preparation, DAX and visualization techniques. The final report connects songs, artists, popularity, chart rankings, explicit content, duration, albums and release trends into a single analytical experience.

Author: Chanchal Khandkure

Role: Aspiring Data Analyst

Skills: Power BI, DAX, SQL, Python, Pandas, NumPy, Excel, Statistics, Data Visualization